Research Methodlogy

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Aditya Kumar

Contents

Chapter 1

Introduction

Understanding what *research* actually means is the first step in building a solid mental model for any project. If you don't get the definition right, you end up just "data-dredging" instead of creating something of value . Think of this as the "OS" of our academic and professional work. Let's break it down into a system you can use.

Explain Like I'm 5

Imagine you have a really tricky puzzle, but some of the pieces are hidden around the house. Research isn't just picking up the pieces you find by accident; it's making a plan to look under every rug and in every drawer until you can put the whole picture together and show your friends how you found it.

The Real Explanation

What: Research is a scientific and systematic search for pertinent information on a specific topic . In a technical sense, it involves defining and redefining problems, formulating hypotheses, and collecting, organising, and evaluating data to reach conclusions .
Why: The core purpose is to find out things in a systematic way to increase knowledge . It aims to discover the truth that is hidden or hasn't been discovered yet . More practically, we do research whenever we gather information to answer a question that solves a problem .
How: It functions as a movement from the known to the unknown—a "voyage of discovery" . This requires a combination of theory and methodology tailored to a specific problem . It must be "*systematic*," meaning it is based on logical relationships rather than just beliefs .
When/Where: It is used whenever a researcher experiences a difficulty in either a theoretical or practical situation and wants a solution . It happens in labs, libraries, the "field," and even in our own homes .

Key Takeaways

1. **Originality:** Research must be an original contribution to the existing stock of knowledge .
2. **Systematic Logic:** It isn't just gathering facts; it's the systematic interpretation of data with a clear purpose .
3. **The Social Contract:** Research is a profoundly social activity that connects you to those who will use your findings to improve their lives or understanding .

MISCONCEPTION ALERT

Misconception: Research is just *intelligence gathering* or *fact-finding* .
Reality: Simply reassembling or reordering facts without interpretation is NOT research . True research requires you to *scratch a mental itch* by answering a *how* or *why* question that moves beyond mere description .

Further Exploration

Deductive vs. Inductive Approaches: Testing existing theories vs. building new ones from data .
The Research Pyramid: Understanding the levels of paradigms, methodology, methods, and techniques .

Cornell notes

Cues / Questions;

1. What is the technical definition of research?
2. How does research differ from "word-spinning"?
3. What is the "voyage of discovery" analogy?
4. Why is "purpose" critical?
5. What is the role of inquisitiveness?

Notes;

1. **Technical Definition:** Redefining problems, formulating hypotheses, collecting/evaluating data, and testing conclusions .
2. **Quality over Quantity:** Research significance lies in its quality, not just "endless word-spinning" .
3. **The Concept of "Systematic":** Based on logical relationships . It requires an explanation of methods and an argument for why results are meaningful .
4. **Academic Activity:** It must be used in a technical sense to involve deductions and reaching conclusions .
5. **The Motivation:** Driven by a "vital instinct of inquisitiveness" to probe the unknown .

Summary; Research is a systematic, goal-oriented process of inquiry that seeks to solve problems or answer questions by contributing new knowledge to an existing field . It is a movement from the known to the unknown, requiring both logical structure and original contribution .

Zettelkastein Notes

RM-001: The Essence of Research Content: Research is "action reading"—a systematic way of structuring thinking and actions to answer a specific question . It is an "art of scientific investigation" that requires the researcher to navigate between the "perceived reality" of problem owners and the "empirical reality" of the data .
Links: RM-002: Methodology vs Methods, RM-003: Problem Definition.
Tags: ResearchFoundations SystematicInquiry KnowledgeCreation

Flow Notes

Problem Indication → Problematising (Interpreting a situation as a problem) → Research Question Formulation → Methodical Search (Data Collection/Analysis) → Conclusion/Solution (New Knowledge) .

Mind Maps

1. Meaning of Research
   (a) Formal Definitions
      i. Systematised effort to gain new knowledge
      ii. Original contribution to knowledge stock
   (b) Key Attributes
      i. Systematic (Logical)
      ii. Objective (Fact-based)
      iii. Goal-Oriented (Problem-solving)
   (c) The Process
      i. Movement: Known → Unknown Cycle: Induction (Searching) & Deduction (Testing)

Chart Comparison

Feature	Mere Information Gathering	True Research
Purpose	Collecting facts with no clear goal	Finding things out systematically to increase knowledge
Structure	Unordered or "pastiche of facts"	Structured, specified steps, logical reasoning
Outcome	Data dump	Original contribution / discovery
Analysis	No interpretation	Data interpreted systematically

Feynman Explanation

Research isn't just about reading a lot of books. It's about being a *detective*. You start with a "mental itch"—a question that bothers you . You look at what everyone else has already said (the literature) so you don't *reinvent the wheel* . Then, you use a specific, logical plan to find new evidence, check if it's true, and explain to others how this new information changes what we know about the world .

Paper Mode

1. Core Idea: Research is a systematic, theory-driven social activity aimed at expanding the boundaries of human knowledge .
2. Findings: It is distinct from mere information collection by its requirement for interpretation, logical consistency, and original contribution .
3. Methodology: It utilizes an "empirical cycle" (inductive or deductive) to move from problem to solution .
4. Contributions: It replaces intuitive decisions with logical ones, serves as the basis for governmental and business policies, and satisfies intellectual curiosity .
5. Limitations: It is often constrained by the researcher's subjectivity, the quality of accessible data, and the "double context" of perceived vs. empirical reality .
6. Discussion: Research is the world's "biggest industry," yet its value depends entirely on the rigour and ethical integrity of the researcher .

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```
# Code Mode
def conductResearch(problem):
question = defineQuestion(problem) # Must be 'how' or 'why' [51]
literature = reviewExistingKnowledge(question) # No reinventing wheels [39]
if literature.fullyAnswers(question):
return "Intelligence Gathering - Not Research" [19]

data = collectSystematically(question) # Methodology choice [52]
findings = interpret(data) # Logic + Reasoning [7]

if findings.isNewContribution(): # Originality check [4]
return findings
else:
```

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return "Refine and Repeat"
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1.1 Objective of Research